

Lecture 16

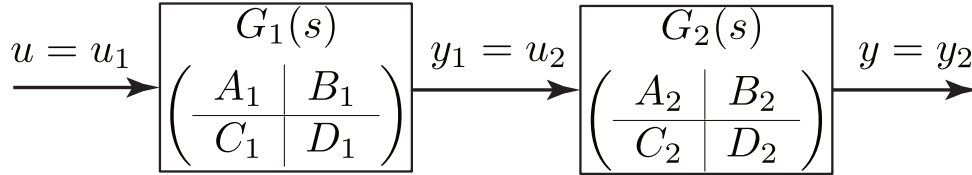
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16.1 Minimality of Interconnected Systems

In this section we shall examine the conditions under which minimality is lost when minimal subsystems are interconnected in various configurations. *When connecting two minimal state-space systems, one might intuitively expect the overall system to remain minimal. However, internal dynamics can become “hidden” from the input or output depending on the structure.*

16.1.1 Series - Cascade Connection

Consider the following system structure where two sub systems, with transfer functions $G_1(s)$ and $G_2(s)$ and associated minimal representations $\left(\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array} \right)$ and $\left(\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array} \right)$, connected in series/cascade configuration.



The transfer function of the connection is simply equal to $G(s) = G_2(s)G_1(s)$. Let x_1 and x_2 be the state-variables of the sub-systems, then the natural choice of the state variable for the series connection is $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Under this definition the state-space representation for the whole system can be found as

$$A = \left[\begin{array}{c|c} A_1 & 0 \\ \hline B_2 C_1 & A_2 \end{array} \right], B = \left[\begin{array}{c} B_1 \\ B_2 D_1 \end{array} \right], C = [D_2 C_1 \mid C_2], D = [D_2 D_1]$$

Clearly eigenvalues of A are the combination of the eigenvalues of A_1 and A_2 and the poles of the system. Let's analyze the observability of the connection via the PBH test.

$$\left[\begin{array}{c|c} \lambda I - A & 0 \\ \hline -B_2 C_1 & \lambda I - A_2 \end{array} \right] = \left[\begin{array}{c|c} \lambda I - A_1 & 0 \\ \hline -B_2 C_1 & \lambda I - A_2 \end{array} \right]$$

If we remember from the observability lecture that (A, C) (whole state-space model) pair is unobservable if and only if $\left[\begin{array}{c|c} \lambda I - A & 0 \\ \hline -B_2 C_1 & \lambda I - A_2 \end{array} \right]$ loses rank for some λ , which can only happen if λ is an eigenvalue of A . Let's assume

that λ_2 is an eigenvalue of A_2 but not eigenvalue of A_1 . Then $\left[\begin{array}{c|c} \lambda I - A & 0 \\ \hline -B_2 C_1 & \lambda I - A_2 \end{array} \right]_{\lambda=\lambda_2}$ loses rank, if $\exists v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ such

that

$$\left[\begin{array}{c|c} \lambda_2 I - A & 0 \\ \hline -B_2 C_1 & \lambda_2 I - A_2 \end{array} \right] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \Rightarrow v_1 = 0 \& \left[\begin{array}{c|c} \lambda_2 I - A & \\ \hline & C_2 \end{array} \right] v_2 = 0$$

which contradicts with the fact that (A_2, C_2) is observable since both individual sub-system representations are minimal. In that respect $\left[\begin{array}{c|c} \lambda I - A & \\ \hline & C \end{array} \right]$ can lose rank only at an eigenvalue of A_1 (i.e. a pole of A_2). Let

λ_1 be an eigenvalue of A_1 . Then $\left[\begin{array}{c|c} \lambda I - A & \\ \hline & C \end{array} \right]_{\lambda=\lambda_1}$ loses rank, if $\exists v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ such that

$$\begin{aligned} \left[\begin{array}{c|c} \lambda_1 I - A & 0 \\ \hline -B_2 C_1 & \lambda_1 I - A_2 \end{array} \right] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= 0 \Rightarrow v_1 \neq 0 \& A_1 v_1 = \lambda_1 v_1 \text{ and} \\ \left[\begin{array}{c|c} \lambda_1 I - A_2 & -B_2 C_1 \\ \hline & D_2 C_1 \end{array} \right] \begin{bmatrix} v_2 \\ v_1 \end{bmatrix} &= 0 \Rightarrow \left[\begin{array}{c|c} \lambda_1 I - A_2 & -B_2 \\ \hline & D_2 \end{array} \right] \begin{bmatrix} v_2 \\ C_1 v_1 \end{bmatrix} = 0 \end{aligned}$$

Note that the block matrix $\left[\begin{array}{c|c} \lambda_1 I - A_2 & -B_2 \\ \hline & D_2 \end{array} \right]$ is the **Rosenbrock System Matrix**. Therefore, the relationship $\left[\begin{array}{c|c} \lambda_1 I - A_2 & -B_2 \\ \hline & D_2 \end{array} \right] \begin{bmatrix} v_2 \\ C_1 v_1 \end{bmatrix} = 0$ implies that λ_1 is a **right zero** of $G_2(s)$ where v_2 and $C_1 v_1$ is the associated *state-zero-direction* and *input-zero-direction* respectively.

Key Takeaway: Observability in Cascade

If we summarize the results, a cascaded system is unobservable if $\exists(\lambda_1, v_1)$ and $v_2 \neq 0$ such that (λ_1, v_1) is an eigenvalue-eigenvector pair of A_1 and λ_1 is a **right zero** of $G_2(s)$ with v_2 and $C_1 v_1$ as the *state-zero-direction* and *input-zero-direction* respectively.

In simpler terms, observability is lost when a pole of the first system is exactly canceled by a transmission zero of the second system.

Now let's analyze the reachability of the connection via the PBH test.

$$\left[\begin{array}{c|c} \lambda I - A & B \end{array} \right] = \left[\begin{array}{c|c|c} \lambda I - A_1 & 0 & B_1 \\ \hline -B_2 C_1 & \lambda I - A_2 & B_2 D_1 \end{array} \right]$$

If we remember from the reachability lecture that (A, B) (whole state-space model) pair is unreachable if and only if $\left[\begin{array}{c|c} \lambda I - A & B \end{array} \right]$ loses rank for some λ , which can only happen if λ is an eigenvalue of A . Let's assume that λ_1 is an eigenvalue of A_1 but not eigenvalue of A_2 . Then $\left[\begin{array}{c|c} \lambda I - A & B \end{array} \right]_{\lambda=\lambda_1}$ loses rank, if

$\exists w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ such that

$$\begin{aligned} \begin{bmatrix} w_1^T & w_2^T \end{bmatrix} \left[\begin{array}{c|c} \lambda_1 I - A & B \end{array} \right] &= 0 \Rightarrow \begin{bmatrix} w_1^T & w_2^T \end{bmatrix} \left[\begin{array}{c|c|c} \lambda_1 I - A_1 & 0 & B_1 \\ \hline -B_2 C_1 & \lambda_1 I - A_2 & B_2 D_1 \end{array} \right] = 0 \\ \Rightarrow w_2^T &= 0 \& w_1^T \left[\begin{array}{c|c} \lambda_1 I - A_1 & B_1 \end{array} \right] = 0 \end{aligned}$$

which contradicts with the fact that (A_1, B_1) is reachable since both individual sub-system representations are minimal. Now let λ_2 be an eigenvalue of A_2 . Then $\left[\begin{array}{c|c} \lambda I - A & B \end{array} \right]_{\lambda=\lambda_2}$ loses rank, if $\exists w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ such

that

$$\begin{aligned}
 \begin{bmatrix} w_1^T & w_2^T \end{bmatrix} \begin{bmatrix} \lambda_2 I - A & B \end{bmatrix} = 0 &\Rightarrow \begin{bmatrix} w_1^T & w_2^T \end{bmatrix} \left[\begin{array}{c|c|c} \lambda_2 I - A_1 & 0 & B_1 \\ \hline -B_2 C_1 & \lambda_2 I - A_2 & B_2 D_1 \end{array} \right] = 0 \\
 \Rightarrow w_2 \neq 0 \ \& \ w_2^T A_2 = w_2^T \lambda_2 \text{ and } \begin{bmatrix} w_1^T & w_2^T \end{bmatrix} \left[\begin{array}{c|c} \lambda_2 I - A_1 & B_1 \\ \hline -B_2 C_1 & B_2 D_1 \end{array} \right] = 0 \Rightarrow \\
 \begin{bmatrix} w_1^T & -(B_2^T w_2)^T \end{bmatrix} \left[\begin{array}{c|c} \lambda_2 I - A_1 & B_1 \\ \hline C_1 & -D_1 \end{array} \right] = 0 &\Rightarrow \begin{bmatrix} w_1^T & -(B_2^T w_2)^T \end{bmatrix} \left[\begin{array}{c|c} \lambda_2 I - A_1 & -B_1 \\ \hline C_1 & D_1 \end{array} \right] = 0
 \end{aligned}$$

Note that $\begin{bmatrix} w_1^T & -(B_2^T w_2)^T \end{bmatrix} \left[\begin{array}{c|c} \lambda_2 I - A_1 & -B_1 \\ \hline C_1 & D_1 \end{array} \right] = 0$ implies that λ_2 is a **left zero** of $G_1(s)$ where w_1 and $(-B_2^T w_2)$ are the associated *state-zero-direction* and *input-zero-direction* respectively.

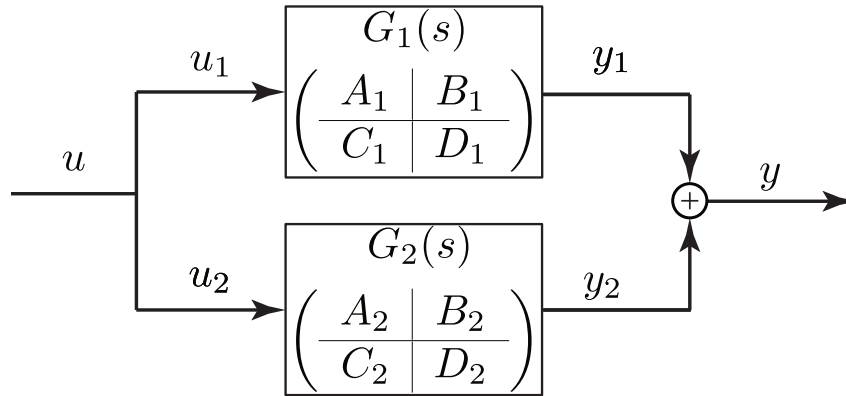
Key Takeaway: Reachability in Cascade

If we summarize the results, a cascaded system is unreachable if $\exists(\lambda_2, w_2)$ and $w_1 \neq 0$ such that (λ_2, w_2) is a left eigenvalue-eigenvector pair of A_2 and λ_2 is a **left zero** of $G_1(s)$ with w_1 and $(-B_2^T w_2)$ as the *state-zero-direction* and *input-zero-direction* respectively.

In simpler terms, reachability is lost when a transmission zero of the first system cancels a pole of the second system.

16.1.2 Parallel Connection

Consider the following system structure where two sub systems, with transfer functions $G_1(s)$ and $G_2(s)$ and associated minimal representations $\left(\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array} \right)$ and $\left(\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array} \right)$, connected in parallel configuration.



The transfer function of the connection is simply equal to $G(s) = G_1(s) + G_2(s)$. Let x_1 and x_2 be the state-variables of the sub-systems, then the natural choice of the state variable for the parallel connection is $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Under this definition the state-space representation for the whole system can be found as

$$A = \left[\begin{array}{c|c} A_1 & 0 \\ \hline 0 & A_2 \end{array} \right], B = \left[\begin{array}{c} B_1 \\ B_2 \end{array} \right], C = [C_1 \mid C_2], D = [D_1 + D_2]$$

Clearly eigenvalues of A are the combination of the eigenvalues of A_1 and A_2 and the poles of the system. Let's analyze the observability of the connection via the PBH test.

$$\left[\begin{array}{c|c} \lambda I - A & 0 \\ \hline C & \end{array} \right] = \left[\begin{array}{c|c} \lambda I - A_1 & 0 \\ \hline 0 & \lambda I - A_2 \\ \hline C_1 & C_2 \end{array} \right]$$

It is easy to observe that combined system is always observable if A_1 and A_2 do not share any common eigenvalue. Thus a necessary (but not sufficient) condition such that parallel connection loses observability is that A_1 and A_2 will have at least 1 common eigenvalue. Let's assume that λ_c is a common eigenvalue/pole of both sub-systems. Then $\left[\begin{array}{c|c} \lambda I - A & 0 \\ \hline C & \end{array} \right]_{\lambda=\lambda_c}$ loses rank, if $\exists v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ such that

$$\left[\begin{array}{c|c} \lambda I - A & 0 \\ \hline C & \end{array} \right] v = 0 \iff \left[\begin{array}{c|c} \lambda I - A_1 & 0 \\ \hline 0 & \lambda I - A_2 \\ \hline C_1 & C_2 \end{array} \right] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \iff$$

$$\begin{aligned} A_1 v_1 &= \lambda_c v_1 \\ A_2 v_2 &= \lambda_c v_2 \\ C_1 v_1 + C_2 v_2 &= 0 \end{aligned}$$

Key Takeaway: Observability in Parallel

The combined system loses observability if and only if $\exists \lambda_c, v_1, \& v_2$ such that λ_c is a common eigenvalue of both systems, and there exists an (v_1, v_2) eigenvector pair (v_i is an eigenvector of A_i associated with λ_c) such that $C_1 v_1 + C_2 v_2 = 0$.

Physically, this means the natural responses of the two subsystems associated with the common pole λ_c completely cancel each other out at the output. The internal modes are excited, but their sum is identically zero.

Now let's analyze the reachability of the connection via the PBH test.

$$\left[\begin{array}{c|c} \lambda I - A & B \end{array} \right] = \left[\begin{array}{c|c|c} \lambda I - A_1 & 0 & B_1 \\ \hline 0 & \lambda I - A_2 & B_2 \end{array} \right]$$

It is easy to observe that combined system is always reachable if A_1 and A_2 do not share any common eigenvalue. Thus a necessary (but not sufficient) condition such that parallel connection loses reachability is that A_1 and A_2 will have at least 1 common eigenvalue. Let's assume that λ_c is a common eigenvalue/pole of both sub-systems. Then $\left[\begin{array}{c|c} \lambda I - A & B \end{array} \right]_{\lambda=\lambda_c}$ loses rank, if $\exists w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ such that

$$w^T \left[\begin{array}{c|c} \lambda_c I - A & B \end{array} \right] = 0 \iff \begin{bmatrix} w_1^T & w_2^T \end{bmatrix} \left[\begin{array}{c|c|c} \lambda_c I - A_1 & 0 & B_1 \\ \hline 0 & \lambda_c I - A_2 & B_2 \end{array} \right] = 0 \iff$$

$$\begin{aligned} w_1^T A_1 &= \lambda_c w_1^T \\ w_2^T A_2 &= \lambda_c w_2^T \\ w_1^T B_1 + w_2^T B_2 &= 0 \end{aligned}$$

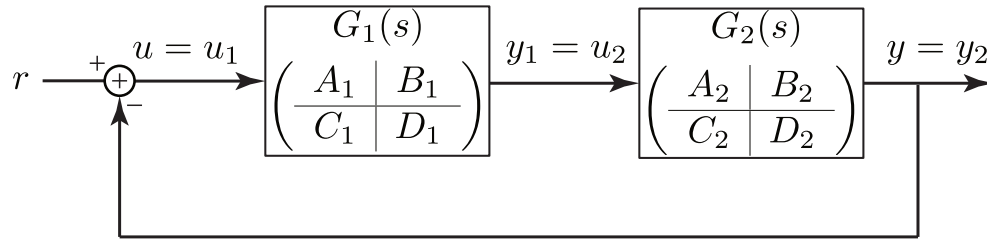
Key Takeaway: Reachability in Parallel

The combined system loses reachability if and only if $\exists \lambda_c, w_1, \& w_2$ such that λ_c is a common eigenvalue of both systems, and there exists an (w_1, w_2) left-eigenvector pair (w_i is a left eigenvector of A_i associated with λ_c) such that $w_1^T B_1 + w_2^T B_2 = 0$.

This implies that the input excites the common mode in both subsystems in equal and opposite ways, resulting in zero net excitation of that mode.

16.1.3 Feedback Connection (Unity)

Consider the following system structure where two sub systems, with transfer functions $G_1(s)$ and $G_2(s)$ and associated minimal representations $\left(\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array} \right)$ and $\left(\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array} \right)$, connected in a unity feedback configuration.



The transfer function of the connection from the reference signal to the output signal is equal to $G(s) = (I + G_1(s)G_2(s))^{-1} G_1(s)G_2(s)$. Let x_1 and x_2 be the state-variables of the sub-systems, then the natural choice of the state variable for the feedback connection is $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Let's also assume that $D_1 = 0$ and $D_2 = 0^1$, under this assumption the state-space representation for the whole system can be found as

$$A_{CL} = \left[\begin{array}{c|c} A_1 & -B_1 C_2 \\ \hline B_2 C_1 & A_2 \end{array} \right], B_{CL} = \left[\begin{array}{c} B_1 \\ 0 \end{array} \right], C_{CL} = [0 \mid C_2], D = [0]$$

Unfortunately it is not straightforward to inspect the eigenvalues of the closed-loop system, since output feedback potentially moves the closed-loop eigenvalue locations. Note that in this topology, the feedforward system is exactly the cascade configuration that we analyzed in the first part. In that respect, we can re-write the closed-loop matrices as

$$\begin{aligned} A_{CL} &= \left[\begin{array}{c|c} A_1 & -B_1 C_2 \\ \hline B_2 C_1 & A_2 \end{array} \right] = \left[\begin{array}{c|c} A_1 & 0 \\ \hline B_2 C_1 & A_2 \end{array} \right] - \left[\begin{array}{c} B_1 \\ 0 \end{array} \right] [0 \mid C_2] = A_{OL} - B_{OL} C_{OL} \\ B_{CL} &= \left[\begin{array}{c} B_1 \\ 0 \end{array} \right] = B_{OL} \\ C_{CL} &= [0 \mid C_2] = C_{OL} \end{aligned}$$

Note that (A_{OL}, B_{OL}, C_{OL}) is the state-space representation of the cascade configuration. Let's assume that

¹Note: We assume $D_1 = D_2 = 0$ to ensure the feedback loop is strictly proper and well-posed, preventing algebraic loops. Even if $D_i \neq 0$, the core results hold as long as $(I + D_1 D_2)$ is invertible.

(A_{CL}, B_{CL}) pair is unreachable, then $\exists w$ such that

$$\begin{aligned} w^T B_{CL} = 0 &\iff w^T B_{OL} = 0 \\ \lambda w^T &= w^T A_{CL} = w^T [A_{OL} - B_{OL} C_{OL}] = w^T A_{OL} \\ \lambda w^T &= w^T A_{OL} \end{aligned}$$

Key Takeaway: Minimality in Feedback Loops

This proves that (A_{CL}, B_{CL}) is unreachable if and only if (A_{OL}, B_{OL}) is unreachable. In that respect, the feedback configuration loses reachability if and only if the cascade configuration in the feedforward path loses reachability.

Similarly, for observability, let's assume that (A_{CL}, C_{CL}) pair is unobservable, then $\exists v$ such that:

$$\begin{aligned} C_{CL} v = 0 &\iff C_{OL} v = 0 \\ \lambda v &= A_{CL} v = [A_{OL} - B_{OL} C_{OL}] v = A_{OL} v \\ \lambda v &= A_{OL} v \end{aligned}$$

This proves that (A_{CL}, C_{CL}) is unobservable if and only if (A_{OL}, C_{OL}) is unobservable. The feedback configuration loses observability if and only if the cascade configuration in the feedforward path loses observability.